

Indian Statistical Institute
M. Math 2nd year
Academic year 2025-2026
Third semester
Midterm Examination
Course: Riemann surfaces
11 - 09 - 25
2 hours

- *Answer as many questions as you can.*
 - *You may use results proved in class, but make sure to state them clearly.*
 - *Maximum marks is 40.*
 - *Unless otherwise mentioned, all Riemann surfaces are assumed to be connected.*
1. Let X be a Riemann surface and let $p \in X$. Let $\mathcal{O}_p$ be the ring of germs of holomorphic functions at p and let $m_p = \{f \in \mathcal{O}_p \mid f(p) = 0\}$. For $f \in \mathcal{O}_p - \{0\}$, define $\text{ord}_p(f) := \text{ord}_{\phi(p)}(f \circ \phi^{-1})$, where ϕ is a chart in a neighbourhood of p .
 - a) Show that $\text{ord}_p(f)$ is well-defined independent of the choice of chart ϕ near p .
 - b) Show that for $\phi \in \mathcal{O}_p$, $m_p = (\phi)$ if and only if $\phi(p) = 0$ and there are neighbourhoods U, V of $p \in X, 0 \in \mathbb{C}$ respectively such that $\phi|_U$ is a biholomorphic map from U to V . (5+5 = 10 marks)
 2. Let $f : X \rightarrow Y$ be a nonconstant holomorphic map between connected Riemann surfaces X, Y .
 - a) Show that f is an open mapping.
 - b) Show that if X is compact, then Y is compact and f is onto.
 - c) Show that any holomorphic function on a compact Riemann surface is constant. (4+3+3 = 10 marks)

3. Let X be a compact Riemann surface. Show that for any $f \in \mathcal{M}(X) - \{0\}$, we have

$$\sum_{p \in X} \text{ord}_p(f) = 0.$$

(8 marks)

4. Let X be a Riemann surface such that $\pi_1(X)$ is a non-trivial cyclic group. Show that X is biholomorphic either to $\mathbb{C}^*$ or $\mathbb{D}^*$ or $A_R = \{1 < |z| < R\}$ for some $R > 1$. (12 marks)

5. Let A be an annulus $A = \{1 < |z| < R\}$ for some $R > 1$. Show that $\text{Aut}(A) = \{z \mapsto \lambda z \mid \lambda \in S^1\} \cup \{z \mapsto \lambda R/z \mid \lambda \in S^1\}$. (10 marks)